Contact-Safe Exploration Without Reinforcement Learning

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Paper under double-blind review

Abstract

Robots often need to explore by touching the world, but contact exploration is where unsafe mistakes are physically costly. Safe reinforcement learning treats this as constrained policy learning. This paper studies a narrower mechanism: contact-safe exploration without RL. The robot grows a certified set of allowed contact probes using local force/displacement limits and only expands the frontier when the certificate margin permits it. A bounded literature sweep over 1436 records places the idea against safe RL, control barrier functions, tactile exploration, active learning, and contact-rich manipulation. In a deterministic toy contact field, random force exploration covers many cells but causes many violations, conservative probing is safe but under-covers, and certificate expansion gives the best safety/coverage tradeoff. The result is not a full benchmark claim. It is a mechanism note: for contact, exploration can be organized around certified physical reachability instead of reward learning.

1 Introduction

Exploration in robotics is not just information gathering. A contact probe can scratch a surface, jam a part, exceed a force limit, or move an object into an unrecoverable configuration. This makes contact exploration different from abstract state-space exploration: unsafe samples are not merely bad data points. They are physical failures.

Safe RL is a powerful framing, but it is not always the right abstraction for contact discovery. Before learning a reward policy, a robot may need to know which touches are allowed at all. We argue for a simpler mechanism: *certificate-guided contact exploration*. The robot maintains a safe set of contact probes, estimates local force limits, and expands only when a guard condition certifies the next probe.

2 Mechanism

Let q index a contact cell and let u be a probe force or displacement. The robot does not know the true local limit $\ell(q)$, but it maintains a conservative estimate $\hat{\ell}(q)$ and a margin m. A probe is allowed when

$$u \leq \hat{\ell}(q) - m.$$

After a successful probe, the robot updates the local estimate and may expand to neighboring contact cells. If the certificate is weak, the robot chooses a lower force or does not expand. This is a safe-set growth rule, not a learned reward policy.

3 Evidence

We evaluate three policies on 2,000 deterministic smooth contact fields. Random force exploration probes aggressively. Conservative exploration uses one low force everywhere. Certificate expansion uses local successful probes to expand a guarded frontier.

The table demonstrates the intended tradeoff. Random exploration obtains coverage by accepting many unsafe contacts. Conservative exploration is safe but does not reach enough

Policy	Avg. covered cells	Violations	Avg. return
Random force	19.675	98684	-46.903
Conservative	14.710	0	2.160
Certificate	26.393	1240	6.384

Table 1: Toy contact-field exploration. Certificate expansion is designed to cover useful contact cells while limiting unsafe probes.

of the contact frontier. Certificate expansion encodes the safety decision directly in the exploration rule.

4 Related Work Boundary

The literature sweep covers safe RL, barrier certificates, model predictive safety filters, tactile exploration, active learning, and contact-rich manipulation. The paper does not claim those areas are missing safety. It claims that contact exploration benefits from an explicit frontier certificate before any reward-learning loop is invoked.

5 Limitations

The evidence is synthetic and uses a hand-written smoothness margin. Real robot contact will require calibrated force sensing, local geometry estimates, and robust treatment of discontinuities. The current paper should be read as a mechanism proposal, not as a deployed exploration system.

6 Conclusion

Contact-safe exploration without RL reframes robot probing as certified safe set growth. For contact-rich robots, the first question is not which action has the best exploration reward, but which action is certified safe to try next.

References

The recovery literature matrix is provided in `docs/related_work_matrix.csv`.